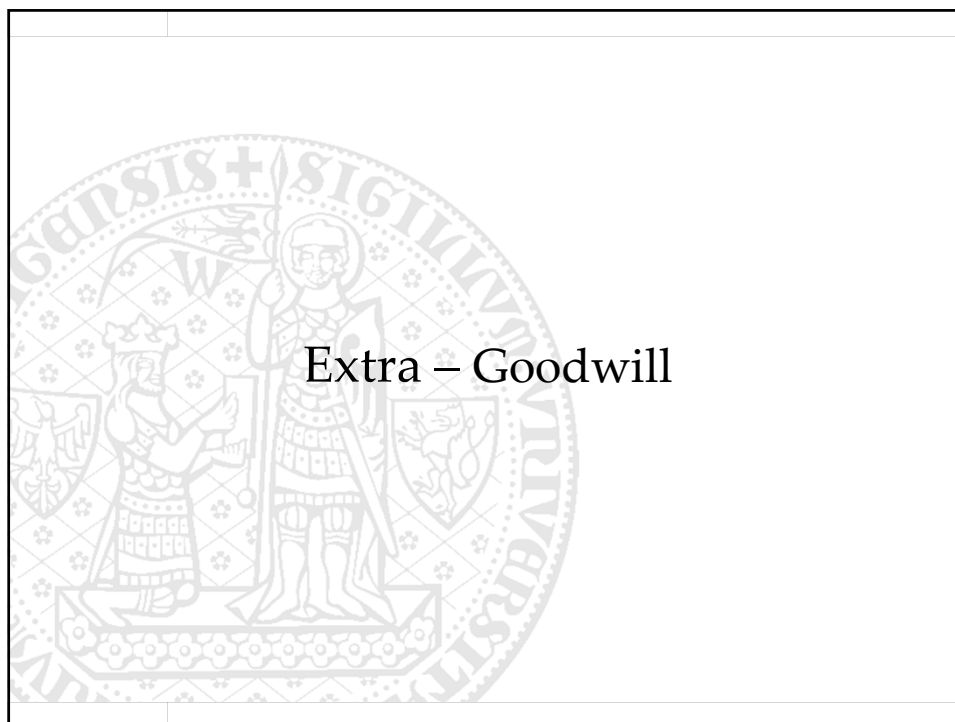






Goodwill

Definition

“Goodwill is an intangible asset that arises as a result of the acquisition of one company by another for a premium value. The value of a company’s brand name, solid customer base, good customer relations, good employee relations and any patents or proprietary technology represent goodwill.”

investopedia.com

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Goodwill

▪ Mergers & Acquisitions

- when BuyerCo acquires a *controlling stake* in TargetCo it becomes its *subsidiary*
- subsidiaries that BuyerCo controls (owns > 50%) are *consolidated*, i.e. their assets and liabilities are reported in BuyerCo's balance sheet
- at acquisition TargetCo's *net assets* ($As - Li$) are revalued through *purchase price allocation*
- $goodwill = \text{purchase price} - \text{revalued net assets}$

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Goodwill

▪ Purchase Price Allocation

- Consider TargetCo with book value of assets of 1,000 revalued to 1,500 and book value of liabilities equal to market value of 300.
- 100% of TargetCo purchased by BuyerCo for 1,700. What *goodwill* does BuyerCo recognize?

TargetCo	book value	revalued
<i>As</i>	1,000	1,500
<i>Li</i>	300	300
<i>Eq</i>	700	1,200

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Goodwill

- **Purchase Price Allocation (TargetCo)**

TargetCo's Balance Sheet

<i>As</i> \$ 1 500	<i>Li</i> \$ 300 <i>Eq</i> \$ 1 700
<i>Gw</i> \$ 500	

▪ $goodwill = 1,700 - (1,500 - 300) = \underline{500}$

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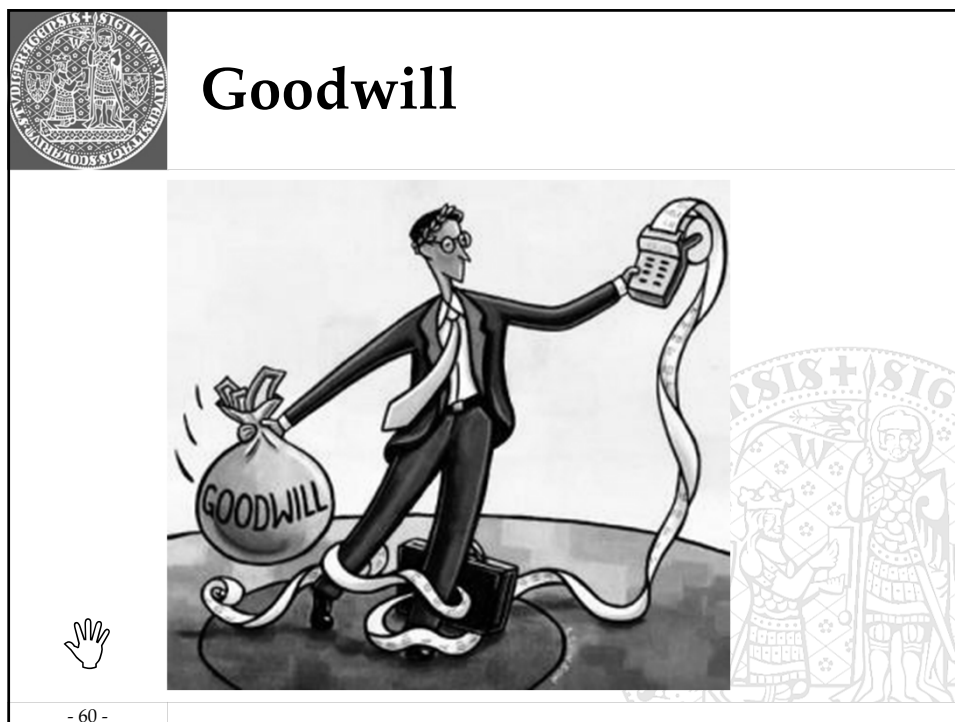


Goodwill

- **Consolidation (BuyerCo)**
 - $\Sigma \text{ debits} = \Sigma \text{ credits}$
 - $1,500 + 500 = 1,700 + 300$

Balance Sheet		Income Statement	
Assets	Liabilities, Equity	Expenses	Revenues
cash			
❶ 1,700			
identifiable assets			
❶ 1,500			
	liabilities		
	❶ 300		
goodwill			
❶ 500			

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Minority Interest

Definition

“Minority interest (or non-controlling interest) is the portion of a subsidiary corporation’s stock that is not owned by the parent corporation. The magnitude of the minority interest in the subsidiary company is generally less than 50% of outstanding shares, otherwise the corporation would generally cease to be a subsidiary of the parent. Minority interest belongs to other investors and is reported on the consolidated balance sheet of the owning company to reflect the claim on assets belonging to other, non-controlling shareholders.”

wikipedia.com

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Balance Sheet

Minority Interest

- SubCo has assets of 1,000 and liabilities of 400.
- 75% of SubCo purchased by MajorCo.
- 25% of SubCo owned by MinorCo.
- Assume market values of assets and liabilities are equal to book values and MajorCo paid no premium over market values to buy the 75% stake.

SubCo	book value	market value
As	1,000	1,000
Li	400	400
Eq	600	600

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Balance Sheet

▪ Minority Interest

1. *no consolidation*

- All three firms report BS separately.



2. *proportional consolidation*

- MajorCo reports 75% of SubCo's *As* and *Li*,
MinorCo reports the remaining 25%



3. *consolidation with minority interest*

- Parent reports all SubCo's *As* and *Li* on its
BS and recognizes MinorCo's *MI* in its *Eq*.



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Balance Sheet

▪ Minority Interest

▪ book value of net assets

- $BV(NA) = As - Li = 1000 - 400 = \underline{600}$
- $BV(75\% NA) = 0.75 \cdot 600 = \underline{450}$

▪ purchase price

- $P(75\% NA) = MV(75\% NA) = \underline{450}$

▪ minority interest

- $MI = 0.25 \cdot BV(NA) = 0.25 \cdot 600 = \underline{150}$

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Balance Sheet

- **Minority Interest**
 - $MI = (\text{revalued net assets} + \text{goodwill}) - \text{purchase price}$

Balance Sheet		Income Statement	
Assets	Liabilities, Equity	Expenses	Revenues
cash			
❶ 450			
identifiable assets			
❶ 1,000			
	liabilities		
	❶ 400		
	minority interest		
	❶ 150		

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Balance Sheet

- **Minority Interest**
 - consolidation with minority is equivalent to:
 - MajorCo raises *Eq* of 150 (25% of 600) of from MinorCo *and*
 - MajorCo uses the 150 together with its own capital of 450 to buy SubCo
- **Treatment in Valuation**
 - minority interest is usually treated as part of equity in computing *ROE*
 - minority claim is then subtracted from firm value before dividing by shares outstanding

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Extra – Goodwill with Minority Interest



Balance Sheet

▪ Minority Interest Based on Book Values

- Acquired SubCo has:
 - *As* – book value of 1,000, revalued to 1,070
 - *Li* – book value equal to market value of 400
- 75% of SubCo purchaed by MajorCo for 570.
- 25% of SubCo owned by MinorCo.

	book value	revalued
<i>As</i>	1,000	1,070
<i>Li</i>	400	400
<i>Eq</i>	600	670

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Balance Sheet

Alternative Approaches

- What would the hypothetical price for 100% stake in SubCo be, i.e. would MinorCo pay premium?

Alternative A

- only MajorCo assumed to pay a premium
- $P(NA) = P(75\% NA) + BV(25\% NA)$
 $= 570 + 0.25 \cdot (1000 - 400)$
 $= \underline{720}$

Alternative B

- MinorCo assumed to pay proportional premium
- $P(NA) = P(75\% NA) / 0.75 = 570 / 0.75 = \underline{760}$

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Balance Sheet

Minority Interest on Book Values

- assumes that *acquisition premium* fully belongs to majority stake
- minority interest computed as MinorCo's *share* in SubCo's *book value* of net assets

$$MI = Pct \cdot BV(NA) = 0.25 \cdot (1,000 - 400) = \underline{150}$$

- goodwill computed as difference between total investment by MajorCo and MinorCo less SubCo's *revalued* net assets

$$Gw = (570 + 150) - (1,070 - 400) = \underline{50}$$

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Balance Sheet

Minority Interest on Book Values

- goodwill = acquisition premium – increase in value of net assets at revaluation

Balance Sheet		Income Statement	
Assets	Liabilities, Equity	Expenses	Revenues
cash			
1 570			
identifiable assets			
1 1,070			
	liabilities		
	1 400		
goodwill	minority interest		
1 50	1 150		

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Balance Sheet

Minority Interest on Book Values

- acquisition premium

$$Prem = (570 + 150) - (1,000 - 400) = \underline{120}$$

- computational equivalence

$$\begin{aligned} Prem &= P_{MajorCo} - BV(75\% NA) = \\ &= 570 - (0.75 \cdot 600) = \\ &= \underline{120} \end{aligned}$$

$$\begin{aligned} Prem &= Reval(NA) + Gw = \\ &= (1,070 - 1,000) + 50 = \\ &= \underline{120} \end{aligned}$$

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Balance Sheet

- **Minority Interest Based on Market Values**

- Acquired SubCo has:
 - *As* – book value of 1,000, revalued to 1,070
 - *Li* – book value equal to market value of 400
- 75% of SubCo purchased by MajorCo for 570.
- 25% of SubCo owned by MinorCo.

	book value	revalued
<i>As</i>	1,000	1,070
<i>Li</i>	400	400
<i>Eq</i>	600	670

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Balance Sheet

- **Minority Interest on Market Values**

- assumes that *acquisition premium* is distributed between the majority and minority stake
- minority interest computed as MinorCo's *share* in SubCo's *market value* of net assets

$$MV(NA) = P_{MajorCo} / 0.75 = 570 / 0.75 = \underline{760}$$

$$MI = MV(NA) \cdot Pct = 760 \cdot 0.25 = \underline{190}$$

- goodwill computed using the higher *MI*

$$Gw = 760 - (1,070 - 400) = \underline{90}$$

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Balance Sheet

- **Minority Interest on Market Values**
 - goodwill = acquisition premium – increase in value of net assets at revaluation

Balance Sheet		Income Statement	
Assets	Liabilities, Equity	Expenses	Revenues
cash			
1 570			
identifiable assets			
1 1,070			
	liabilities		
	1 400		
goodwill	minority interest		
1 90	1 190		

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Balance Sheet

- **Minority Interest on Market Values**
 - acquisition premium

$$Prem = (570 + 150) - (1,000 - 400) = \underline{120}$$
 - implied premium for 100% NA

$$Impl = 120 / 0.75 = \underline{160}$$
 - computational equivalence

$$\begin{aligned}
 Impl &= (Reval(NA) - BV(NA)) + Gw = \\
 &= (1,070 - 1,000) + 90 = \\
 &= \underline{160}
 \end{aligned}$$

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